

Experiment 1

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Aim

Getting started with the Python 3.x and installing libraries of Tensorflow, Keras, Pytorch.

Platform

Anaconda Navigator

Theory

Anaconda is an open-source distribution for python and R. It is used for data science, machine learning, deep learning, etc. With the availability of more than 300 libraries for data science, it becomes fairly optimal for any programmer to work on anaconda for data science. Anaconda helps in simplified package management and deployment. Anaconda comes with a wide variety of tools to easily collect data from various sources using various machine learning and AI algorithms. It helps in getting an easily manageable environment setup which can deploy any project with the click of a single button.

Procedure

Installation of Anaconda

1. Visit <https://www.anaconda.com/products/individual>
2. Download the installer.
3. Launch the installer
4. Choose whether to add Anaconda to your PATH environment variable. We recommend not adding Anaconda to the PATH environment variable, since

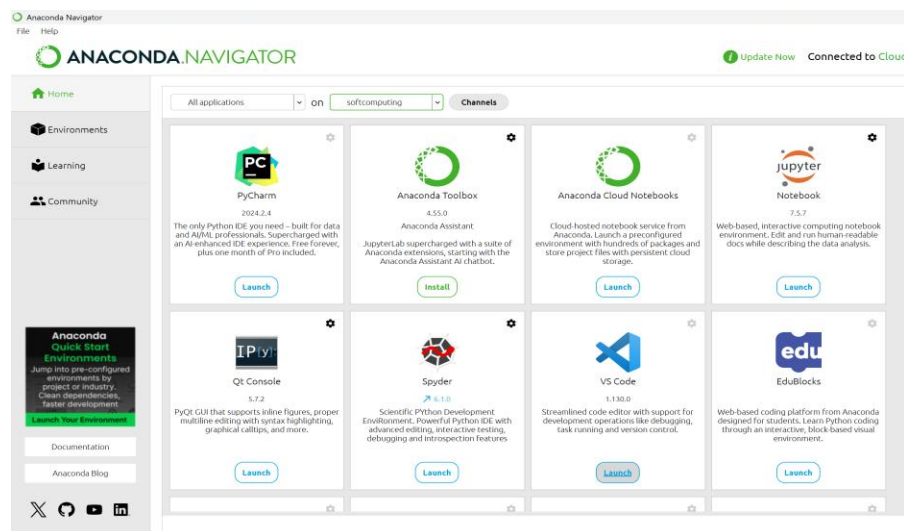
this can interfere with other software. Instead, use Anaconda software by opening Anaconda Navigator or the Anaconda Prompt from the Start Menu.

5. Choose whether to register Anaconda as your default Python. Unless you plan on installing and running multiple versions of Anaconda or multiple versions of Python, accept the default and leave this box checked.
6. Click the Install button. If you want to watch the packages Anaconda is installing, click Show Details.
7. Click the Next button.
8. After a successful installation you will see the “Thanks for installing Anaconda” dialog box.
9. If you wish to read more about Anaconda.org and how to get started with Anaconda, check the boxes “Anaconda Individual Edition Tutorial” and “Learn more about Anaconda”. Click the Finish button.

Installing libraries of Tensorflow, Keras, Pytorch

1. Keras, TensorFlow and PyTorch are among the top three frameworks that are preferred by Data Scientists as well as beginners in the field of Deep Learning. This comparison on Keras vs TensorFlow vs PyTorch will provide you with a crisp knowledge about the top Deep Learning Frameworks and help you find out which one is suitable for you.
2. Open Anaconda Navigator.
3. Type TensorFlow in the search box, select the TensorFlow and apply to install the compatible TensorFlow package.
4. Now type in Keras and install Keras library to our environment.
5. Install other libraries as and when required.

Screenshots



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Clone

Import

Backup

Remove

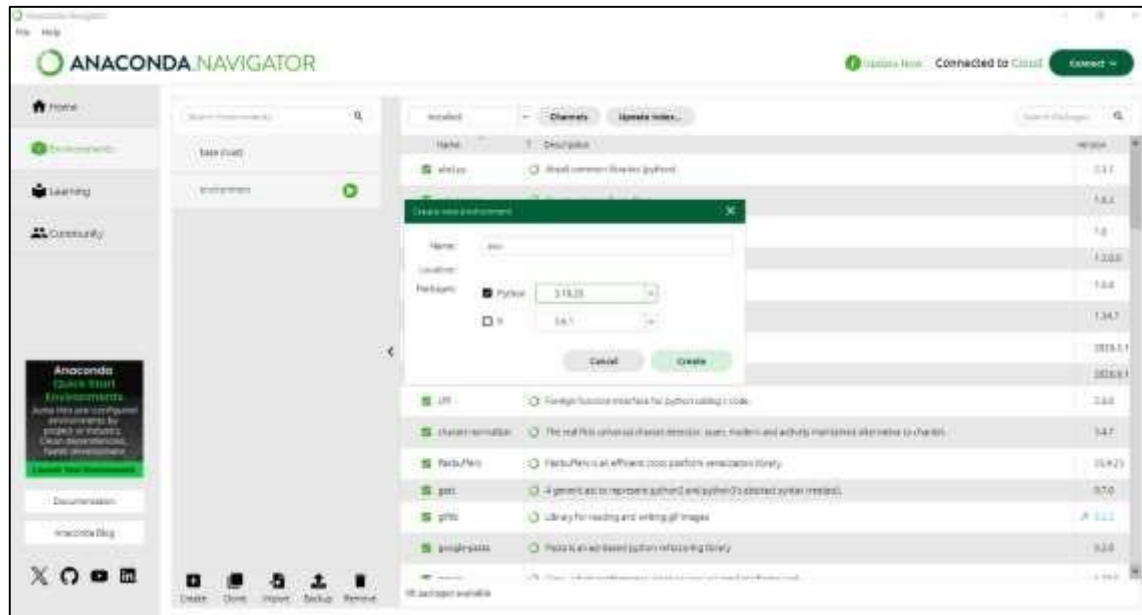
✓

async-lru

○

Simple Iru_cache for asyncio

315 packages available



Install Packages

×

The following packages will be modified:

Solving package specifications

Cancel

Apply

Results

Anaconda Navigator successfully installed. Libraries TensorFlow, Keras, and PyTorch also installed.

Learning Outcomes

- Setting up and configuring Anaconda for the projects
- Creation of the environment using Anaconda Navigator
- Installation of various libraries like PyTorch, TensorFlow, and Keras